Compute Standard Errors for a Fitted sjSDM Model

Description

Post-hoc calculation of standard errors for a fitted sjSDM model using 'sjSDM::getSe()'. This is intentionally separated from 'fit_jsdm_model()' so that the GPU-based model fitting step and the CPU-based SE computation step can be run independently, allowing the SE step to exploit all available CPU cores without being constrained by the GPU device setting.

Usage

```
compute_jsdm_se(

  mod_jsdm = NULL,

  parallel = 0L,

  step_size = NULL,

  verbose = FALSE

)
```

Arguments

`mod_jsdm`	An object of class 'sjSDM' returned by 'fit_jsdm_model()' or 'sjSDM::sjSDM()'. Must not be 'NULL'.
`parallel`	Number of CPU cores to use for the data loader during SE computation. Passed to the 'parallel' argument of 'sjSDM::getSe()'. Default is '0L' (no parallelisation). Use the 'n_cores' value from 'config.model_fitting' in the project configuration to exploit all available cores.
`step_size`	Batch size for stochastic gradient descent used during SE computation. Passed to 'sjSDM::getSe()'. Default is 'NULL', which lets sjSDM choose automatically.
`verbose`	Logical scalar. If 'FALSE' (default), all console output from 'sjSDM::getSe()' — including Python/reticulate stdout — is suppressed. Set to 'TRUE' to see progress messages, which is recommended when running inside a pipeline target to monitor long-running SE computation.

Details

'sjSDM::getSe()' uses CPU for SE computation regardless of the device used for model fitting. Separating SE computation into its own pipeline target therefore allows the caller to pass the full number of available CPU cores via 'parallel' without conflict with the GPU device setting used during fitting.

When 'verbose = FALSE', output is captured with 'reticulate::py_capture_output()' (Python stdout) and 'utils::capture.output()' (R stdout) so the console remains clean. When 'verbose = TRUE', both output streams are printed for monitoring progress.

If SE computation fails, the function raises an error with an informative message.

Value

The input 'mod_jsdm' object with its 'se' field populated with the computed standard errors.

See Also

'sjSDM::getSe', 'fit_jsdm_model'